

4(a)	any four from: 1 (put) extract / sample / concentrate / pigments, on, pencil / base, line ; 2 (named) solvent below, extract / line, and idea of time to rise / left till solvent rises / wait till solvent almost reaches top ; 3 remove paper (before solvent reaches end) and mark solvent front ; 4 ($R_f =$) distance pigment moved $\div$ distance solvent moved ; 5 check / compare with, known / reference / published, data / values ;	4 2 e.g. ethanol / acetone / petroleum ether $\pm$ hexane / trichloromethane 2 DO NOT ALLOW water 5 ALLOW database / standard values
4(b)	any three from: 1 absorb <u>light</u> ; 2 for, light dependent (stage) / photophosphorylation / photoactivation ; 3 absorb / use, more / different, wavelengths OR (different pigments have) different / extend, absorption spectra ; 4 AVP ;	3 IGNORE colours throughout 1 ALLOW harvest / capture / trap, for 'absorb' 2 IGNORE photolysis 2 CON if process described with error e.g. pigment passes electrons to another pigment 4 e.g. chlorophyll a is, primary pigment / reaction centre and, carotene / xanthophyll / chlorophyll b, are accessory pigments e.g. (some pigments may) protect from excess light